

Time and Space complexities and Arrays

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constraints

$$\leq 400$$

$$\leq 10^4$$

$$\leq 10^5$$

$$\leq 10^8$$

$$\leq 10^9$$

Time complexity

$$O(n^3)$$

$$O(n^2)$$

$$O(n \log n)$$

$$O(n)$$

$$O(\log n)$$

Cutoff $\rightarrow \underline{10^8}$

$$n \leq \text{arr.length} \leq \underline{10^4}$$

$$O(n)$$

$$n = 10^4$$

$$\rightarrow \boxed{10^4 \leq 10^8}$$

$$O(n^2)$$

$$10^8 \leq 10^8$$

$$O(n^3)$$

$$10^{12} \times$$

$$O(n^3) \quad \checkmark$$

$$10^{12} \quad \times$$

$$----- 10^8 -----$$

$$\text{nums.length} = 10^5$$

$$n = 10^5$$

$$O(n)$$

$$\boxed{O(n^2)}$$

$$n \log n$$

$$10^5 \times \log_2 10^5$$

$$[0, 1, 1, 1, 2, 2, 3, 3, 4]$$